

Exercise 1: Reservoir Engineering

Problem 1

a) Fluid Distribution and Water Saturation in an Anticlinal Reservoir

An anticline forms a structural trap when porous reservoir rock is overlain by impermeable caprock. Gravity segregation places the least dense phase at the top: gas accumulates at the crest, oil lies below the gas-oil contact (GOC), and the densest phase, water, occupies the lowest part of the structure below the oil-water contact (OWC).

In a homogeneous, water-wet reservoir, water saturation increases downward. It is highest in the water zone, at and below the free-water level, where $S_w \approx 1$. It is lowest near the crest, where capillary pressure is greatest and S_w approaches irreducible water saturation, S_{wi} . The gradual change between these regions is the capillary transition zone.

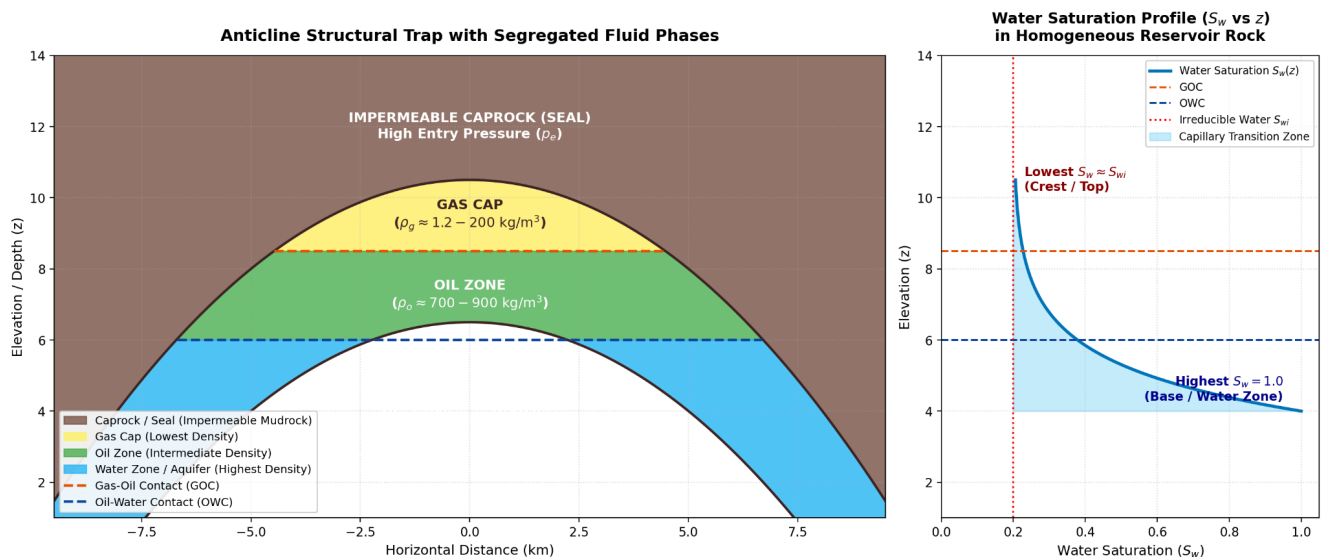


Figure 1. Digital illustration of an anticlinal structural trap showing gravity segregation of gas, oil, and water beneath an impermeable caprock. The accompanying simplified profile shows water saturation increasing downward from irreducible water saturation near the crest to approximately 100% in the water zone.

b) Effect of Grain Size and Sorting on Porosity

For a given grain shape and packing arrangement, absolute grain size does not affect porosity. If all grains are scaled equally, pore volume and bulk volume change by the same proportion, so $\phi = V_p/V_b$ remains unchanged.

Grain-size distribution has a strong effect. Well-sorted sediments, containing grains of similar size, retain larger interconnected pore spaces and therefore have higher porosity. Poorly sorted sediments contain smaller grains that fill the pores between larger grains, reducing pore volume and porosity. For example, if large-grain packing has $\phi = 0.30$, and its pores are filled with a material having $\phi = 0.30$, the combined porosity may decrease to $0.30 \times 0.30 = 0.09$.

c) Total and Effective Porosity

Total porosity, ϕ_t , is the ratio of all pore volume, including connected and isolated pores, to bulk rock volume:

$$\phi_t = \frac{V_{p,total}}{V_b}$$

Effective porosity, ϕ_e , is the fraction of bulk volume occupied by interconnected pores that can contain and transmit fluids:

$$\phi_e = \frac{V_{p,connected}}{V_b}$$

The distinction is important because only effective porosity contributes to fluid flow and recoverable hydrocarbon volume. Using total porosity where isolated pores or clay-bound microporosity are significant may overestimate storage capacity and reserves. In clean, well-connected sandstones, ϕ_e is often close to ϕ_t ; the difference can be substantial in shales, cemented rocks, and some carbonates.

Effective porosity is commonly measured by helium porosimetry or by vacuum saturating a dry core with brine and measuring the gained fluid volume. Total porosity can be determined from the bulk volume of the intact core and the true grain volume measured after crushing or disaggregating the sample; this includes pores that were originally isolated.

d) Effective Porosity from Water Saturation

The saturated plug has a larger uncertainty because water may remain on the outer surface after wiping, may evaporate during handling, or may drain from surface pores. These effects change the measured mass, whereas the dry plug has a stable, repeatable weight.

The bulk volume of the cylindrical plug is

$$V_b = \frac{\pi}{4} d^2 h = \frac{\pi}{4} (2.20)^2 (3.87) = 14.71 \text{ cm}^3$$

Its relative uncertainty is

$$\frac{\Delta V_b}{V_b} = \sqrt{\left(2 \frac{0.02}{2.20}\right)^2 + \left(\frac{0.02}{3.87}\right)^2} = 0.0189$$

$$V_b = 14.71 \pm 0.28 \text{ cm}^3$$

The water-filled pore volume is

$$V_p = \frac{m_{sat} - m_{dry}}{\rho_w} = \frac{32.5 - 27.5}{1.0} = 5.00 \text{ cm}^3$$

$$\Delta V_p = \frac{\sqrt{(0.1)^2 + (0.0)^2}}{1.0} = 0.10 \text{ cm}^3$$

Thus, the effective porosity is

$$\Phi_e = \frac{V_p}{V_b} = \frac{5.00}{14.71} = 0.340$$

$$\frac{\Delta \Phi_e}{\Phi_e} = \sqrt{\left(\frac{0.10}{5.00}\right)^2 + \left(\frac{0.28}{14.71}\right)^2} = 0.0275$$

$$\phi_c = 0.340 \pm 0.009$$

This is effective porosity because only pores accessible to the saturating water are measured.

Problem 2 — Boyle's-Law Helium Porosimetry

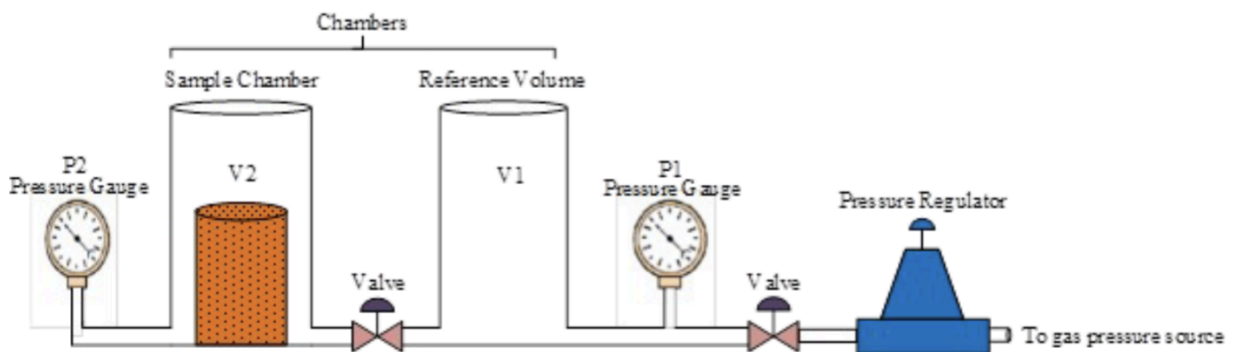


Figure 2. Schematic of a Boyle's-law helium porosimeter. A core plug is placed in the sample chamber (V_2), which is connected to a reference volume (V_1). The initial pressures, P_1 and P_2 , are measured before opening the isolation valve; after equilibration, the common pressure P_3 is used to determine the matrix volume and porosity of the core plug.

a) Matrix-volume equation

Assume the plug is placed in chamber 1. Before opening the valve, the gas volumes are $V_1 - V_m$ in chamber 1 and V_2 in chamber 2. After equilibration, the total gas volume is $V_1 + V_2 - V_m$.

Conservation of gas mass at constant temperature gives

$$p_1(V_1 - V_m) + p_2V_2 = p_3(V_1 + V_2 - V_m)$$

Rearranging for matrix volume:

$$V_m = \frac{p_3(V_1 + V_2) - p_1V_1 - p_2V_2}{p_3 - p_1}$$

b) Calculation

$$V_m = \frac{7.5(50+50) - 2(50) - 10(50)}{7.5 - 2} = 27.27 \text{ cm}^3$$

The bulk volume of the cylindrical plug is

$$V_b = \frac{\pi}{4} d^2 L = \frac{\pi}{4} (3.0)^2 (4.5) = 31.81 \text{ cm}^3$$

$$V_p = V_b - V_m = 31.81 - 27.27 = 4.54 \text{ cm}^3$$

$$\phi = \frac{V_p}{V_b} = \frac{31.81 - 27.27}{31.81} = 0.143$$

$$V_m = 27.27 \text{ cm}^3$$

$$\phi = 0.143 = 14,3\%$$